

# Nitin M

FULL-STACK DEVELOPER · AI/ML ENGINEER · GAME DEVELOPER

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## I. ABOUT ME

Curious, highly inquisitive, and hands-on. I naturally want to understand how things work beneath the surface. I am not satisfied with simply knowing what something does — I want to know why it works, how it works, and whether there is a better way to build it. I learn by questioning, experimenting, breaking things, rebuilding them, and turning ideas into working projects. When something interests me, I can become deeply involved in it, often going beyond what is required simply because I want to understand the subject properly. This has led me to explore full-stack development, AI/ML, cybersecurity, data systems, software architecture, and game development.

I am naturally a problem solver and systems thinker. I enjoy taking a complicated problem, breaking it into smaller pieces, understanding how those pieces interact, and then building a solution from the ground up. I also enjoy connecting ideas from different fields and using whatever technology is appropriate rather than restricting myself to one stack.

Creativity is another important part of how I work. I enjoy building things that are not just functional but interesting — from environmental intelligence and AI scheduling platforms to cybersecurity systems and a complete narrative game with its own mechanics, story and systems.

I value independence and ownership. I like figuring things out for myself, taking responsibility for the systems I build, documenting what I learn, and continuously improving my work. At the same time, I enjoy collaborative environments where people can challenge ideas, share knowledge, and build something better together.

Most importantly, I am driven by curiosity rather than a checklist of technologies. I want to become the kind of engineer who can face an unfamiliar problem, learn what is necessary, design a solution, and actually build it. I build with AI-assisted development as a core part of my workflow — writing specs, orchestrating coding agents and reviewing what they produce.

**In short:** I am a builder, a question-asker, a problem solver, and someone who genuinely enjoys learning by creating.

## II. SKILLS & EXPERTISE

### Programming

Python · JavaScript · TypeScript · C++ · SQL · HTML/CSS

### Backend

Node.js · Express.js · Flask · REST APIs · JWT · OAuth 2.0

**Databases** MongoDB · Mongoose · Prisma · SQLite · Redis

**Tools** Git/GitHub · Postman · Blender · Cron Jobs · Vercel

### Frontend

React · Next.js · Vite · Tailwind CSS · Framer Motion · Recharts

### AI & ML

Scikit-Learn · XGBoost · SMOTE · K-Means · DBSCAN · Isolation Forest

**Game Dev** Unreal Engine 5 · Blueprints · Pygame

## III. FEATURED PROJECTS



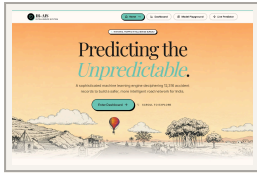
### FULL-STACK PLATFORM WORK IN PROGRESS

## EcoSense India

*An environmental intelligence platform that unifies weather, AQI and satellite data for any place in India.*

- Combines 10 data sources: ground stations, forecast models, satellites and reanalysis
- Express / MongoDB backend with scheduled collection, forecast and backfill jobs
- JWT + Google sign-in, map search down to locality level, 12h → 30d trend dashboards

React · Node.js · Express · MongoDB · MapTiler · Recharts  
[github.com/Nitinshakthi7/Ecosense](https://github.com/Nitinshakthi7/Ecosense)



MACHINE LEARNING

## IR-AIS

*India Road Accident Intelligence System: predicts accident severity and segments high-risk conditions from 12,316 records.*

- Slight / Serious / Fatal classification with 26 engineered features
- SMOTE for class imbalance; PCA + K-Means risk segmentation
- Interactive Next.js predictor and model-comparison dashboard

Python · Scikit-Learn · XGBoost · Next.js · TypeScript  
[github.com/Nitinshakthi7/IR-AIS-India-Road-Accident-Intelligence-System](https://github.com/Nitinshakthi7/IR-AIS-India-Road-Accident-Intelligence-System)



CYBERSECURITY · ML

## Behavioral Threat Detection

*An unsupervised framework that surfaces suspicious network behaviour without labelled attacks.*

- Analysed 175K+ UNSW-NB15 network sessions
- DBSCAN + Isolation Forest with anomaly-consensus scoring
- Apriori association rules to explain behavioural patterns

Python · Scikit-Learn · K-Means · DBSCAN · Isolation Forest · Apriori  
[github.com/Nitinshakthi7/Network-Behavior-and-Cybersecurity-Pattern-Analysis-Using-Unsupervised-Learning](https://github.com/Nitinshakthi7/Network-Behavior-and-Cybersecurity-Pattern-Analysis-Using-Unsupervised-Learning)



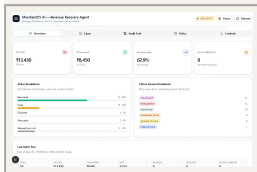
GAME DEVELOPMENT IN DEVELOPMENT

## Looped Lies

*A solo-developed 15–20 hour narrative action-adventure in Unreal Engine 5, built around a time loop.*

- 16,500-word story across five acts with three endings
- Stealth, branching dialogue, moral alignment and save/load systems
- 1,341-line technical development guide

Unreal Engine 5 · C++ · Blueprints · Nanite · Lumen



HACKATHON · FINTECH AGENT RAZORPAY BUILDATHON

## MerchantOS

*A policy-bounded agent loop that recovers failed payments end to end, with a full audit trail. Ran a 50-case simulated batch: 62.9% recovered.*

- Five-stage agent loop: Detect → Diagnose → Decide → Act → Verify, every step audited
- Swappable Razorpay clients: the official SDK, or a deterministic in-memory simulator
- Live dashboard: KPIs, recovery cases, audit trail and run controls

Next.js · React · TypeScript · Prisma · SQLite · Tailwind CSS · Recharts  
[github.com/Nitinshakthi7/MerchantOS](https://github.com/Nitinshakthi7/MerchantOS)



GAME · FULL-STACK TEAM OF 4

## MysteryDesk

*A full-stack detective investigation game with five cases, branching interrogations and a server-derived contradiction engine — built with a four-person team.*

- Team of four: I rebuilt a one-case prototype into the full five-case game
- Contradictions between testimony and evidence computed at query time; the answer key is never stored
- Five endings derived from the player's actual investigation

React · Vite · Node.js · Express · SQLite · GSAP  
[github.com/TejashRajuKV/Mystery\\_Desk](https://github.com/TejashRajuKV/Mystery_Desk)

## IV. MORE PROJECTS

HACKATHON · OPTIMISATION

### TimeGenius.AI

CICADA HACKATHON

*Conflict-free university timetables for 50+ courses in under 3 seconds, with PDF/Excel export.*

Flask · OR-Tools · React · SQLite

TOOL · LEVEL DESIGN

### KOSMOS

*A level editor for building engine-ready interior spaces and reusable parametric assets, exported as deterministic JSON.*

React · TypeScript · Vite · Gemini API

AlgoViz

Step through 59 algorithms and data-structure operations, with animated visuals and the code running line by line.

JavaScript · HTML · CSS · SVG

Used Car Price & Insurance

Six-algorithm comparison over 2,805 records:  $R^2$  0.883 on price, 81% on insurance status.

Python · Scikit-Learn · Pandas · GridSearchCV

Don't Even Bother

A merciless troll platformer: 19 levels, two difficulty modes, a solver bot to race, everything generated at runtime. Live on itch.io.

Python · Pygame · NumPy · OOP

Snake & Ladder Quiz COURSE FINAL PROJECT

A vanilla-JS board game with quiz-gated movement and true-random dice from an external API.

JavaScript · HTML/CSS · Open Trivia DB · Random.org

GameVault

A personal game library with a collection, a wishlist and filters, backed by a secured REST API (15+ endpoints).

Express · MongoDB · JWT · HTML/CSS/JS

Smart Energy Tracker

Home → Room → Device → Reading hierarchy with authenticated REST APIs and cost analytics.

Node.js · Express · MongoDB · JWT

Station Quest

A text-based escape room built in Figma Make: six rooms, a natural-language command parser, a 22-step optimal escape.

React · TypeScript · Vite · Figma Make

Kanban Board COURSE PROJECT

A full-featured board built entirely from prompts: drag-and-drop, roles, activity log and auth — no shallow scaffold.

Next.js · React · Prisma · NextAuth.js

V. CAREER DIRECTION

**What I want:** opportunities where I can work on real products, difficult engineering problems and technology with practical impact. I am especially interested in full-stack product engineering, AI/ML applications, backend systems, data-driven products, automation and game technology.

**Where I want to grow:** teams that value curiosity, experimentation, engineering depth and ownership — where I can learn from strong engineers while taking responsibility for meaningful parts of a product.

**Companies I aspire to work toward:** Google · Microsoft · NVIDIA · OpenAI · Amazon · Apple · Adobe · Epic Games · Unity and other high-impact technology/product companies.

"Take a problem from idea → architecture → implementation → intelligent system → finished product."

VI. EDUCATION

Atria University — Bengaluru

2024 – 2028 (Expected) · B.Tech Digital Transformation · Minor: AI/ML

New Horizon Pre-University College

2024 · PUC / 12th Grade — 60%

SJR Public School — Bengaluru

2022 · 10th Grade — 65%

VII. STRENGTHS

- Systems Thinking
- Problem Solving
- Rapid Prototyping
- Technical Documentation
- Creative Thinking
- Collaborative Development
- Learning Through Building